2-25-26 (WEEK 1)

ALGEBRAIC INTEGERS

DEFINITION

K is a number field if $K \supseteq \mathbb{Q}$ is a finite extension of fields

↳ Intuition: $\mathbb{Q} = \mathrm{Frac}(\mathbb{Z})$, $K = \mathrm{Frac}(?)$

↖ replace $\mathbb{Z}$

DEFINITION

$\alpha \in K$ is an algebraic integer if a [monic poly $f \in \mathbb{Z}[x]$ exists s.t. $f(x) = 0$] (Δ)

EQUIVALENT DEFINITION

$\alpha \in K$ is an alg int if min poly of α over $\mathbb{Q}$ is with coef $\in \mathbb{Z}$

<u>Proof</u>

Let f satisfy (Δ) with min deg f.

$\mathbb{Z}$: ED $\Rightarrow$ f: irred in $\mathbb{Z}[x]$

$\therefore$ By Gauss Lemma, f: irred in $\mathbb{Q}[x]$ $\square$

DEFINITION

$\mathcal{O}_K := \{\alpha \in K \mid \alpha$: alg int, i.e. is integral $/\mathbb{Z}\}$

GOAL

① $\mathcal{O}_K$ is a ring

② $\mathcal{O}_K$ is a fin gen $\mathbb{Z}$-mod

INTEGRALITY

We assume here rings are comm w/ unit

Also, we have $A \subseteq B$ as an extension of rings

DEFINITION

(i) $\alpha \in B$, then α is integral over A if $\exists$ monic $f(x) \in A[x]$, s.t. $f(x) = 0$ (guarantee if f: irred)

(ii) We say B is integral over A if $\forall \alpha \in B$, α is integral over A

PROPOSITION

For $x \in B$, TFAE:

(i) $x \in B$ is integral over A — main point

(ii) $A[x] \subseteq B$ is a f.g. A-module — module: we don't consider multiplication, unlike w/ algebra (and yet still f.g.)

(iii) $\exists A[x] \subseteq A' \subseteq B$ extension of rings s.t. A' is a f.g. A-module

<u>Proof</u>

(i) $\Rightarrow$ (ii) $\Rightarrow$ (iii): Trivial

(iii) $\Rightarrow$ (i): $\exists \alpha_i \in A'$, s.t. $A' = \{\sum_{i=1}^{n} a_i \alpha_i \mid a_i \in A\}$

Define the A-linear map $m_x : A' \to A$

$a' \mapsto x \cdot a'$

$\therefore A' \ni x \cdot \alpha_i = \sum_{j=1}^{n} a_{ij} \alpha_j \ \forall i$

$$\Rightarrow \begin{pmatrix} x & & \\ & \ddots & \\ & & x \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} = (a_{ij}) \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} \Rightarrow \underbrace{[x I_n - \overbrace{(a_{ij})}^{M}]}\begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} = 0$$

$$\therefore \underbrace{M^* M}_{= \det(M) I_n} \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} = 0 \Rightarrow \det(M) \alpha_i = 0 \ \forall i \underset{1 \in A}{\Rightarrow} \det(M) = 0$$, i.e. $\exists$ monic poly f s.t. $f(x) = 0$ $\square$

COROLLARY + DEFINITION

Define $A' := \{x \in B \mid x \text{ integral over } A\}$

Then, A' is a ring and we call A' the integral closure of A in B.

Proof

Claim: $\forall x, y \in A'$, we have $x+y,\ xy \in A'$

Proof: By prop, $A[x,y]$ is f.g. $A[x]$-mod and $A[x]$ is f.g. A-mod $\square$

In particular, $\mathcal{O}_K$ is a ring

EXERCISE

Let d be a squarefree integer and $K := \mathbb{Q}(\sqrt{d})$

Then, $\mathcal{O}_K := \begin{cases} \mathbb{Z} \oplus \mathbb{Z}\frac{\sqrt{d}+1}{2}, & d \equiv 1 \pmod 4 \\ \mathbb{Z} \oplus \mathbb{Z}\sqrt{d}, & d \equiv 2, 3 \pmod 4 \end{cases}$

COROLLARY

TFAE

(i) B is integral over A and is a f.g. A-algebra

(ii) B is a f.g. A-module

Proof

(i) $\Rightarrow$ (ii): Proposition ✓

(ii) $\Rightarrow$ (i): By induction on # generators of B/A + proposition $\square$

COROLLARY

Consider the ring extension $A \subseteq B \subseteq C$, if B: integral over A + C: integral over B, then C: integral over A

Proof

$\alpha \in C \Rightarrow \alpha^n + b_{n-1}\alpha^{n-1} + \dots + b_0 = 0$ for some $b_i \in B$

$\therefore B' = A[b_0, \dots, b_{n-1}]$: integral over A and f.g. A-alg

$\Rightarrow$ By coro, B' is a f.g. A-mod

α: int $/B' \Rightarrow B'[\alpha]$: f.g. B'-mod $\Rightarrow$ f.g. A-mod $\square$

DEFINITION

(i) For $A \subseteq B$, we say A is integrally closed in B if the integral closure of A in B is in A itself

(ii) An integral domain is integrally closed if the integral closure of A in $\mathrm{Frac}(A)$ is A itself

└ fraction field of A

LEMMA

Let $S \subseteq A \setminus \{0\}$ be a multiplicative set

(e.g. ① $P \trianglelefteq A$: prime ideal, $S = A \setminus P$: multiplicative set

② $f \in A$: not n.potent, i.e. $f^n \neq 0\ \forall n \in \mathbb{Z}_{\geq 0}$, then take $S = \{f^n\}$)

If we define $S^{-1}A := \{\frac{a}{s} \mid \frac{a_1}{s_1} = \frac{a_2}{s_2} \Leftrightarrow \exists u \in S, \text{ s.t. } (a_1 s_2 - a_2 s_1)u = 0\}$

① $S = \{f^n\} \Rightarrow f$: invertible in $S^{-1}A$

② $S = A \setminus P \Rightarrow S^{-1}A$: unique max ideal $P(S^{-1}A)$

Now, let A' be an integral closure of A in B, then prove $S^{-1}A'$ is the integral closure of $S^{-1}A$ in $S^{-1}B$.

(exercise)

PROPOSITION

UFD is integrally closed

Proof

A: UFD, $K = \mathrm{Frac}(A)$.

Then, $\forall \alpha \in K \setminus \{0\}$, $\alpha = \frac{a}{b}$: int over A (a, b w/o common irred factors)

Choose a monic $f \in A[T]$ w/ min deg s.t. $f(\alpha) = 0$. In particular, f: irred in $K(x)$

Thus, $\alpha^n + a_{n-1}\alpha^{n-1} + \dots + a_0 = 0$. As f: irr, thus $a_0 \neq 0$

$\therefore a^n + \boxed{a_{n-1}a^{n-1}b + \dots + a_0 b^n} = 0$, so $b \mid a^n$ $\times$

DEFINITION

$ab \in P \Rightarrow a \in P$ or $b \in P$

$\mathrm{Spec}\, A = \{$all prime ideals of $A\}$

LEMMA

Let $A \subseteq B$ be an integral extension of rings

Let $P \subseteq Q$ be prime ideals in B, s.t. $P \cap A = Q \cap A$. Then, $P = Q$

Proof

Let $p = P \cap A = Q \cap A \Rightarrow A/p \hookrightarrow B/p$

We reduce to the case where $p = 0$ and A, B: integral domains

If $Q \neq 0$, choose $x \in Q \setminus \{0\}$.

As B int/A, $\exists$ monic $f(T) \in A[T]$ with min deg $f \Rightarrow f$: irr in $A[T]$

$\therefore x^n + a_{n-1}x^{n-1} + \dots + a_0 = 0$ with $a_0 \neq 0$

Then, $a_0 \in (x) \cap A \subseteq Q \cap A = p = 0$ $\times$

PROPOSITION

Let $A \subseteq B$ be an integral ext of rings.

(1) Suppose B is a domain. Then, B: field $\Leftrightarrow A$: field

(2) $P \triangleleft B$ is a prime ideal ($P \in \mathrm{Spec}(B)$). Then, P: max ideal of $B \Leftrightarrow p = P \cap A$: max ideal of A

Proof

It suffices to show (1)

① A is a field

Fix $x \in B \setminus \{0\}$. Then, $\exists$ monic irred $f(T) \in A[T]$, s.t. $f(x) = 0$

(exercise)